Chapter 5

Hyperbolic Problems

In Chapter 2 we have introduced the heat equation. Using the classification scheme of Appendix B.1, we can identify the heat equation as a *parabolic* equation. We now move on to consider other types, and begin with hyperbolic problems.

5.1 The Wave Equation

We consider the wave equation as a pure initial value problem (a *Cauchy Problem*):

$$\begin{aligned} u_{tt} &= c^2 u_{xx}, \qquad x \in \mathbb{R},\ t > 0, && (5.1)\\ u(x,0) &= \phi(x), \qquad x \in \mathbb{R}, && (5.2)\\ u_t(x,0) &= \psi(x), \qquad x \in \mathbb{R}. && (5.3) \end{aligned}$$

Note the the wave equation has a second derivative in time, in contrast to the heat equation where only the first derivative in time appears. Hence, we need initial conditions for both the initial state and the initial time derivative.

In Appendix B.3 we have shown that the general solution to the Cauchy problem (5.1)-(5.3) is given by

$$u(x,t) = \frac{1}{2}\left(\phi(x-ct) + \phi(x+ct)\right) + \frac{1}{2c}\int_{x-ct}^{x+ct} \psi(\xi)\, d\xi.$$

From the solution formula it is clear that the wave equation has finite speed of propagation. The solution at any location (x,t) is only affected by the initial conditions in the interval $[x-ct, x+ct]$. Conversely, the initial conditions at any x_0 can only influence a region bounded by the straight lines $x_0 \pm ct = const$. These lines are called *characteristics*. It should also be noted, that the regularity of the solution at (x,t) is now determined by the regularity of the initial conditions. The solution is not smooth, unless both ϕ and ψ are. This should be contrasted to the properties of the heat equation outlined in section 1.3.1.

5.2 Hyperbolic Systems

We will not work with the wave equation directly, but rather identify another equation that is hidden (so to speak) in the wave equation. Start with

$$u_{tt} = c^2 u_{xx},$$

and set, assuming a smooth solution,

$$\begin{aligned} v := u_t \quad &\Longrightarrow v_t = u_{tt} && (5.4)\\ w := u_x \quad &\Longrightarrow w_t = u_{xt} = u_{tx} = v_x. && (5.5) \end{aligned}$$

We can re-write (5.4) using the wave equation, and (5.5),

$$v_t = u_{tt} = c^2 u_{xx} = c^2 w_x.$$

We can thus transform the wave equation into a system of *first-order* partial differential equations in v and w:

$$\begin{pmatrix} v_t \\ w_t \end{pmatrix} = \begin{pmatrix} 0 & c^2 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} v_x \\ w_x \end{pmatrix},$$

or, shorthand, using $\mathbf{v} = (v, w)^T$,

$$\mathbf{v}_t + A^{(w)}\mathbf{v}_x = 0. \tag{5.6}$$

Here w stands for "wave equation", and

$$A^{(w)} = \begin{pmatrix} 0 & -c^2 \\ -1 & 0 \end{pmatrix}.$$

The eigenvalues of $A^{(w)}$ are given by $\lambda_{1,2} = \pm c$, and the corresponding eigenvectors are given by $\mathbf{r}_{1,2} = (\mp c, 1)^T$. The characteristics of the wave equation are thus found in the eigensystem of $A^{(w)}$. We are ready to define a general hyperbolic system:

Definition 5.1 *A linear system of m first-order PDE*

$$\mathbf{v}_t + A\mathbf{v}_x = 0 \tag{5.7}$$

is called hyperbolic*, if $A \in \mathbb{R}^{m\times m}$ is diagonalizable, and has only real eigenvalues.*

We will substantially generalize this definition later, in order to admit nonlinear multidimensional systems as well. But for now this definition will suffice. As an example, we can decompose the hyperbolic system (5.6), for which $A = A^{(w)}$, as follows. Let $\Lambda = diag(\lambda_1, \lambda_2)$ be a diagonal matrix containing the two eigenvalues. Furthermore, let $R = (\mathbf{r}_1|\mathbf{r}_2)^T$ be a matrix containing as columns the two eigenvectors. Then we have

$$0 = \mathbf{v}_t + A^{(w)}\mathbf{v}_x = \mathbf{v}_t + R\Lambda R^{-1}\mathbf{v}_x$$

We multiply with R^{-1} from the left to obtain

$$R^{-1}\mathbf{v}_t + \Lambda R^{-1}\mathbf{v}_x = \mathbf{u}_t + \Lambda\mathbf{u}_x = 0,$$

where $\mathbf{u} := R^{-1}\mathbf{v}$ has been defined. Note that the equations are decoupled, as Λ is a diagonal matrix! Let $\mathbf{u} = (u^{(1)}, u^{(2)})^T$. Then we have two *scalar* equations of the form

$$u_t^{(i)} + \lambda_i u_x^{(i)} = 0, \qquad i = 1, 2.$$

A similar decomposition can be done for all systems which fall under Definition 5.1. Apparently, the core of such hyperbolic systems is of the type $u_t + au_x = 0$, where a can be any real constant. We shall consider such problems next.

5.3 The Advection Equation

We have motivated in the previous section that at the root of linear hyperbolic equations there lies an equation even simpler than the wave equation, i.e. the advection equation

$$u_t + au_x = 0. \tag{5.8}$$

Here $a \in \mathbb{R}$ is a constant. (Note that a can be positive or negative!) The advection equation has a straightforward interpretation. Note that (5.8) is obtained, if we define the gradient $\nabla u := (u_t, u_x)^T$, and $\mathbf{d} = (1, a)^T$, and require

$$\nabla u \cdot \mathbf{d} = u_t + a u_x = 0. \tag{5.9}$$

Hence the advection equation can be thought of as the statement that the gradient of u vanishes in direction $(1, a)^T$. This simple structure makes it particularly easy to give a well-posedness result at least for sufficiently regular solutions of the advection equation. For conciseness, we consider the following model problem:

Model Problem 3 Let $u_0 : C^1(\mathbb{R}) \to \mathbb{R}$, and assume that u_0 is bounded, with bounded first derivatives. Find $u \in C^1(\mathbb{R} \times [0, \infty))$, such that, for given $a \in \mathbb{R}$,

$$u_t + a u_x = 0, \qquad x \in \mathbb{R},\ t > 0, \tag{5.10}$$

$$u(x, 0) = u_0(x), \qquad x \in \mathbb{R}. \tag{5.11}$$

Note that we have assumed that the initial conditions (and its first derivative) are bounded. This will make the following proof easier. We will later extend the solution to larger classes of functions.

Theorem 5.1 *Model Problem 3 is well-posed.*

Proof By direct differentiation, one can check that for any given C^1 function u_0, the C^1 function $u : \mathbb{R} \times [0, \infty) \to \mathbb{R}$, defined by

$$u(x, t) = u_0(x - at) \tag{5.12}$$

solves the advection equation. It is obvious that this function also satisfies the initial conditions at $t = 0$. Conversely, by the transport property (5.9), we see that the function defined by $v(s) := u(x + as, t + s)$, where u is any C^1-solution of (5.10), is invariant in s. (Compute v' and use the advection equation (5.10).) This immediately implies that $u(x, t) = v(0) = v(-t) = u(x - at, 0)$. By the initial conditions, we see that any C^1 solution must necessarily be given by (5.12).

The solution formula (5.12) also directly implies the maximum principle

$$\max_{(x,t) \in \mathbb{R} \times (0,\infty)} |u(x, t)| = \max_{x \in \mathbb{R}} |u_0(x)|.$$

Furthermore, straightforward differentiation of (5.12) shows that the derivatives of u can be bounded by u_0', with a constant that depends only on a. In summary then, if u, v are two solutions for initial conditions u_0, v_0, respectively, then by linearity $u - v$ solves (5.10), (5.11), with initial conditions $u_0 - v_0$, and we have

$$\|u - v\|_{C^1(\mathbb{R} \times (0,\infty))} \le C \|u_0 - v_0\|_{C^1(\mathbb{R})},$$

where the constant C depends only on a, and $\|\cdot\|_{C^1(\mathbb{R} \times (0,\infty))}$ is a suitable norm that makes the restriction of C^1 to bounded functions a Banach space, e.g. $\|u_0\|_{C^1(\mathbb{R})} = \max_{x \in \mathbb{R}} |u_0(x)| + \max_{x \in \mathbb{R}} |u_0'(x)|$ (similarly for u). ∎

The defining characteristic of the advection equation is the simple transport property (5.9), which says that the solution is constant along any straight line, where $x - at$ is constant: The solution is "transported" along the direction $(1, a)^T$, and we call the corresponding family of straight lines *characteristics*.

It is easy to see that if we are dealing with a finite domain, the characteristics determine appropriate boundary conditions. Let $a > 0$, then the solution is propagated from "left to right" along the characteristic lines. Assume that we want to solve the advection equation with appropriate boundary conditions for $(x, t) \in \overline{\Omega} := [0, 1] \times [0, T]$. In a manner entirely analogous to that of Theorem 5.1, one can show that a well-posed formulation with Dirichlet boundary conditions is given by

$$u_t + au_x = 0, \qquad (x, t) \in \Omega_T := (0, 1] \times (0, T] \tag{5.13}$$

$$u(x, 0) = u_0(x), \qquad x \in [0, 1], \tag{5.14}$$

$$u(0, t) = g(t), \qquad t \in (0, T]. \tag{5.15}$$

Note that we cannot freely prescribe boundary conditions at $x = 1$, because the PDE says that the solution there is determined by propagation of the data. Figure 5.1 shows an illustration of the computational domain. Note that for $a < 0$, the situation is opposite: We cannot prescribe boundary data at $x = 0$, but conversely must give boundary data at $x = 1$.

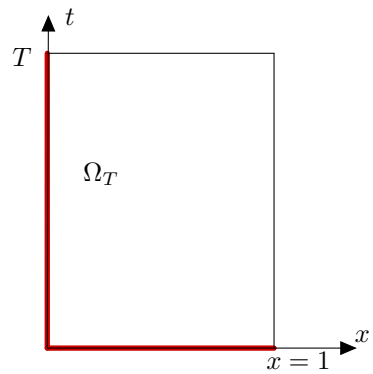

Figure 5.1: Computational domain of Problem (5.13)–(5.15). We have highlighted the boundary segments $\Gamma := \overline{\Omega} \backslash \Omega_T$, to indicate where we prescribe data for the case $a > 0$.

5.4 Further Reading

In Evans [5] the advection equation is discussed in Section 2.1, and the wave equation is discussed in Section 2.4. The general solution we have derived for the one-dimensional case is given there, but also the higher dimensional case is covered. Furthermore, solution formulae for problems with boundary conditions, and the well-posedness of the wave equation are discussed. Strauss [12] has similar things on the wave equation (basics are discussed in section 2-4, and chapter 9 discusses the higher dimensional case). As usual, this text may be a bit more accessible.

Chapter 6

Approximation of the Advection Equation

6.1 A Simple Finite Difference Scheme

By Taylor Series we establish easily

$$\frac{u(x,t+\tau)-u(x,t)}{\tau} = u_t(x,t) + \frac{1}{2}\tau u_{tt}(x,t) + \mathcal{O}(\tau^2), \tag{6.1}$$

$$\frac{u(x+h,t)-u(x-h,t)}{2h} = u_x(x,t) + \frac{1}{6}h^2 u_{xxx}(x,t) + \mathcal{O}(h^4). \tag{6.2}$$

Just like we did for the heat equation, we have thus defined a first-order approximation in time, and a second order central approximation in space. The latter approximation is now for the first derivative. In Section 3.5 we established that such a type of approximation was stable under a time-step restriction for the heat equation, and we may hope to get a similar result here. For simplicity, let us consider again the periodic problem. We define the spatial grid as in (4.4), and the temporal grid as in (4.5). Denote by u_i^n the numerical solution at point $(x_i, t^n) \in \mathcal{G}_{h,\tau} := \mathcal{G}_h \times \mathcal{G}_\tau$, and combine (6.1) and (6.2) to produce an approximation for the advection equation:

$$\frac{u_i^{n+1}-u_i^n}{\tau} + a\frac{u_{i+1}^n - u_{i-1}^n}{2h} = 0. \tag{6.3}$$

Adding the initial conditions, and periodic boundary conditions, we can define the following scheme:

Scheme 5 (Central Scheme for the Advection Equation) For $i = 0, 2, \ldots, N-1$ and $n = 0, 1, 2, \ldots$ set

$$\begin{aligned} u_i^{n+1} &= u_i^n - \frac{\nu}{2}\left(u_{i+1}^n - u_{i-1}^n\right), \\ u_i^0 &= u_0(ih) \\ u_{i+N}^n &= u_i^n \end{aligned} \tag{6.4}$$

where $\nu = \frac{a\tau}{h}$. Here h and τ are the grid parameters corresponding to the grids $\mathcal{G}_h$ (cf. (4.4)), and $\mathcal{G}_\tau$ (cf. (4.5)), respectively.

We need to investigate consistency and stability of the scheme: If the scheme is both consistent and stable, the Lax-Equivalence principle tells us that it will be convergent. The consistency

error is defined in Def. 3.1, where eq. (3.20) states that we ought to substitute the exact solution of the advection equation into the discrete scheme, and collect error terms. From the Taylor series (6.1) and (6.2), we obtain immediately that the scheme is consistent with the advection equation,

$$\frac{u(x,t+\tau)-u(x,t)}{\tau}+a\frac{u(x+h,t)-u(x-h,t)}{2h}=\underbrace{u_t+au_x}_{=0}+\frac{\tau}{2}u_{tt}+\mathcal{O}(\tau^2+h^2), \tag{6.5}$$

with a consistency error of or $\mathcal{O}(\tau+h^2)$. The scheme is second-order consistent if we require $\tau=\mathcal{O}(h^2)$.

To investigate stability, we consider von-Neumann analysis. The amplification factor defined in Section 4.4 reads

$$|\rho_k(h,\tau)|=\left|1-\frac{\nu}{2}\left(e^{\imath kh}-e^{-\imath kh}\right)\right|, \qquad k=0,\dots N-1.$$

Using $e^{\pm\imath kh}=\cos(kh)\pm\imath\sin(kh)$, we have, setting $\xi_k:=kh$,

$$|\rho_k(h,\tau)|=|1-\imath\nu\sin(\xi_k)|=\sqrt{1+\nu^2\sin^2(\xi_k)}\geq 1 \qquad \forall k,$$

with strict inequality for the maximum taken over k, and any $\nu>0$. It thus becomes clear that we do not meet the von-Neumann condition[1]. We have to re-examine the discretization.

How can we explain that the discretization does not work? Insight can be gained by looking at the so-called modified equation:

Theorem 6.1 *The central scheme (Scheme 5) is consistent of order $\mathcal{O}(\tau^2+h^2)$ with the equation*

$$u_t+au_x=-\frac{\tau a^2}{2}u_{xx} \tag{6.6}$$

Proof Let u solve (6.6), and substitute u into the Central Scheme. We obtain:

$$\begin{aligned}\frac{u(x,t+\tau)-u(x,t)}{\tau}+a\frac{u(x+h,t)-u(x-h,t)}{2h}&=u_t+au_x+\frac{\tau}{2}u_{tt}+\mathcal{O}(\tau^2+h^2)\\&=-\frac{\tau a^2}{2}u_{xx}+\frac{\tau}{2}u_{tt}+\mathcal{O}(\tau^2+h^2)\end{aligned} \tag{6.7}$$

Note that for u a smooth solution of (6.6), we can write

$$\begin{aligned}u_t &= -au_x+\mathcal{O}(\tau)\\ \Leftrightarrow u_{tt} &= -au_{xt}+\mathcal{O}(\tau)\\ &= -au_{tx}+\mathcal{O}(\tau)\\ &= a^2u_{xx}+\mathcal{O}(\tau).\end{aligned}$$

Substituting this into (6.7) yields

$$\begin{aligned}\frac{u(x,t+\tau)-u(x,t)}{\tau}+a\frac{u(x+h,t)-u(x-h,t)}{2h} &= -\frac{\tau a^2}{2}u_{xx}+\frac{a^2\tau}{2}u_{xx}+\mathcal{O}(\tau^2+h^2)\\ &= \mathcal{O}(\tau^2+h^2)\end{aligned}$$

■

[1] If we set $\tau=\mathcal{O}(h^2)$, we can meet at least the weaker statement $|\rho_k|\leq 1+C\tau$ (cf. Remark 4.1). This can be seen by noting that $\sin^2(\xi)\leq 1$, and doing a Taylor series expansion in ν for $\sqrt{1+\nu^2}$. However, here we opt for the stricter condition $|\rho_k|\leq 1$, as we have seen that the analytical solution of the advection equation is a simple transport of the initial conditions, and thus we should not permit any growth.

It may seem surprising that the central scheme is *more* accurate for a PDE which we never intended to approximate. However, considering the nature of the difference approximation, and that of the error when expressed in terms of a Taylor series expansion, this is actually not unusual. In fact, every difference approximation is more accurate for some equation, other than the one we intend to approximate, if some of the terms remaining in the Taylor series are included in the equation. The modified equation can help to characterize the behavior of the scheme. Here we see that the right-hand side of (6.6) is given by negative diffusion, a term that also appears when solving the heat equation backwards in time, cf. Example 2.1. Recall that this leads to exponential growth in the solution. It is thus not entirely surprising that the central difference approximation is only stable is the sense of $r \leq 1 + C\tau$, which permits exponential growth (cf. Remark 4.1).